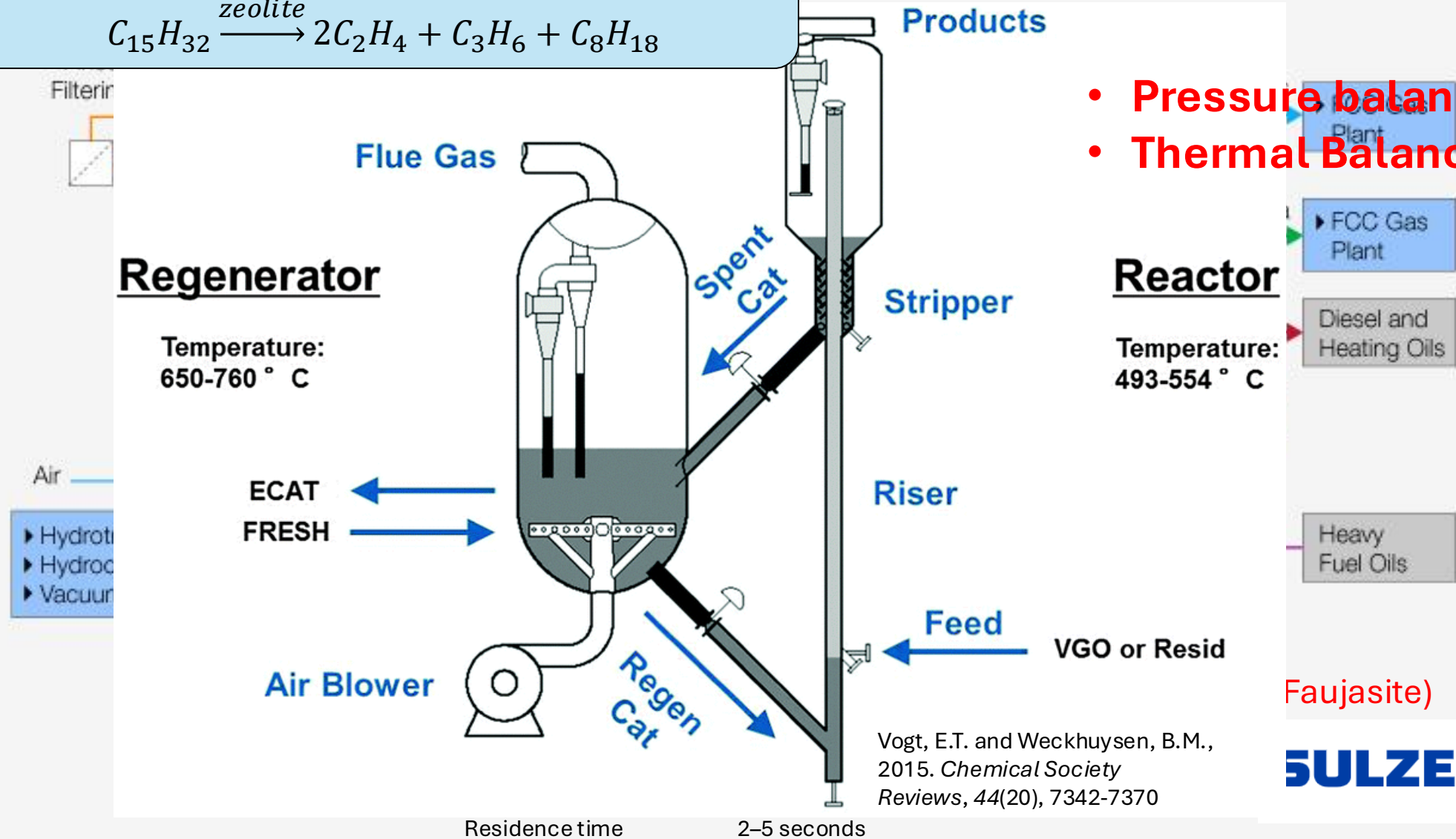
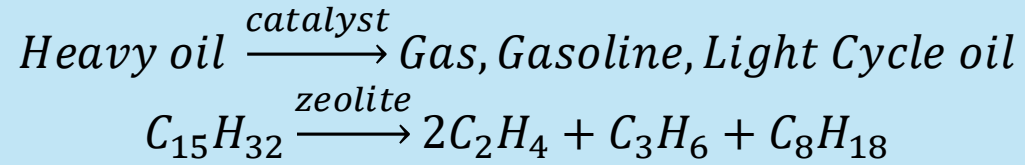


Chemical Reaction Engineering – II (CL24303)

MO2026

Fluidized Catalytic Cracking (FCC)

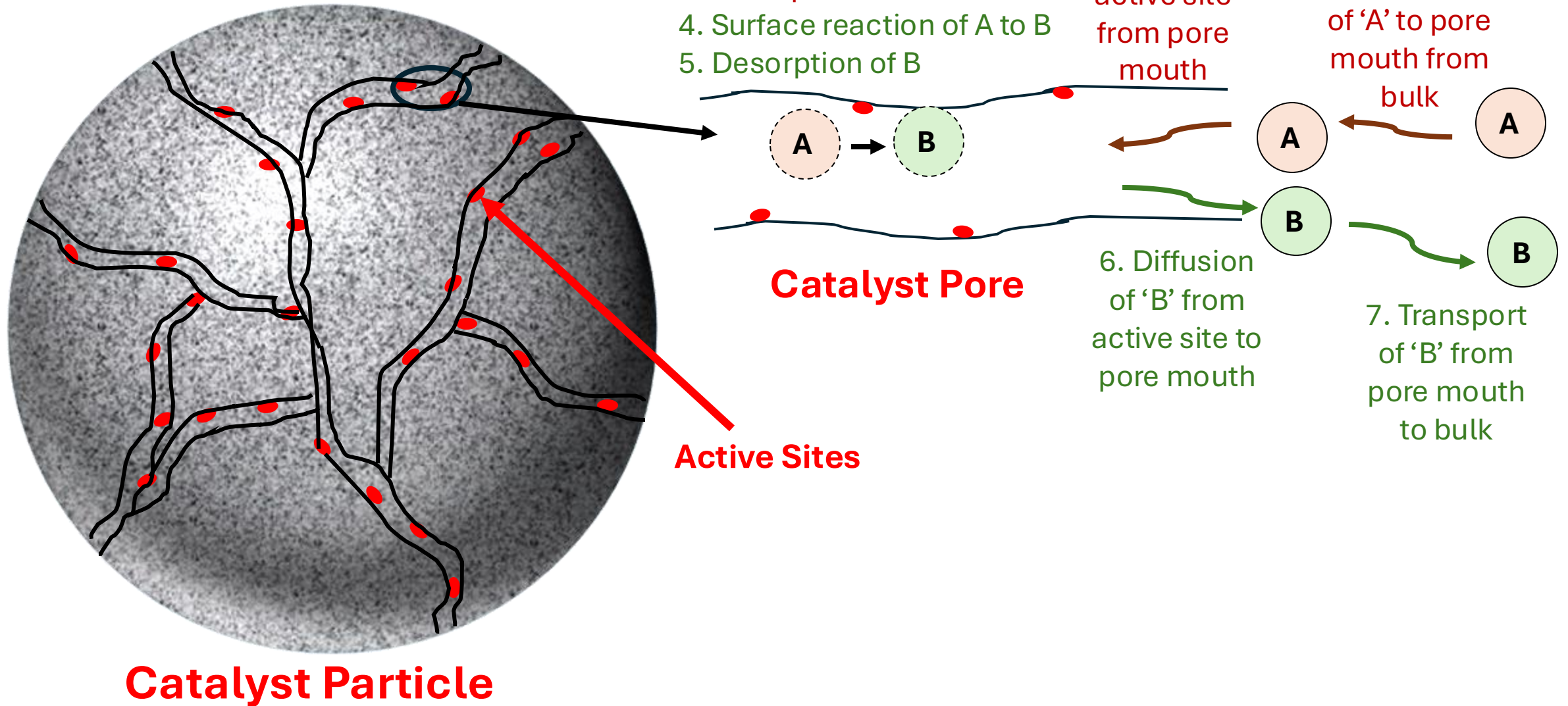


- Pressure balance
- Thermal Balance

Vogt, E.T. and Weckhuysen, B.M.,
2015. *Chemical Society
Reviews*, 44(20), 7342-7370

SULZER

Steps in Catalytic Reaction



Catalyst Shapes



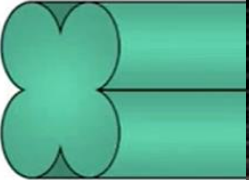
Shape Key

- 1 Honeycomb monolith (cylinder)
- 2 Honeycomb monolith (disc)
- 3 Gear-shaped extrudate
- 4 Multi-hole support block
- 5 Wagon-wheel / spoked tablet
- 6 Rings (hollow cylinders)
- 7 Multi-hole extrudate (small)
- 8 Spherical pellets / beads
- 9 Cylindrical extrudates (tablets)
- 10 Lobed extrudates (tri/quadrilobe)
- 11 Broken extrudate rods
- 12 Cubic / hexagonal tablets

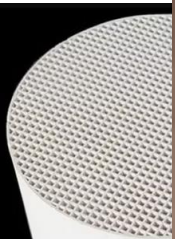
Size	Use
2-10 mm	packed tubular reactors
> 1 mm	packed tubular reactors,
1-20 mm	Packed tubular reactors, moving beds
8-14 to 2-4 mesh, >2 mm	packed tubular reactors
	liquid phase reactors
20-300 μm	fluidized reactors
75-200 μm	slurry reactors



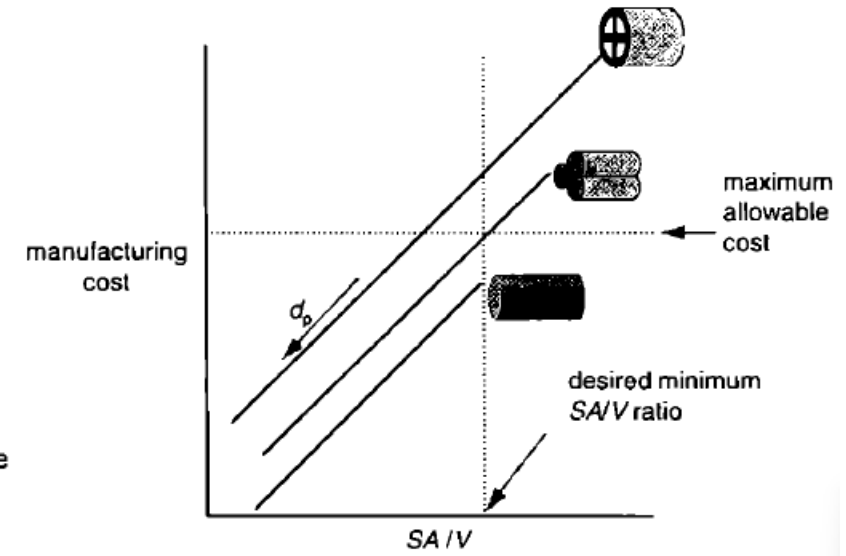
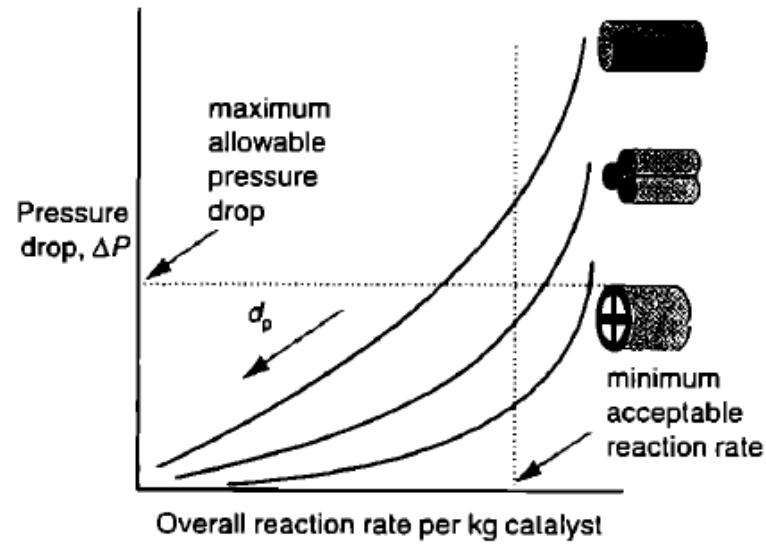
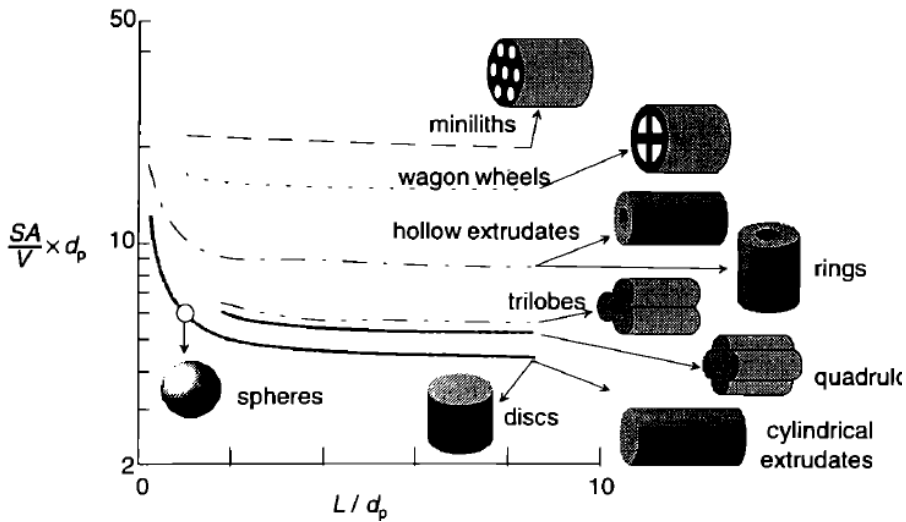
Sphere



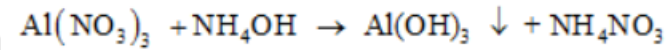
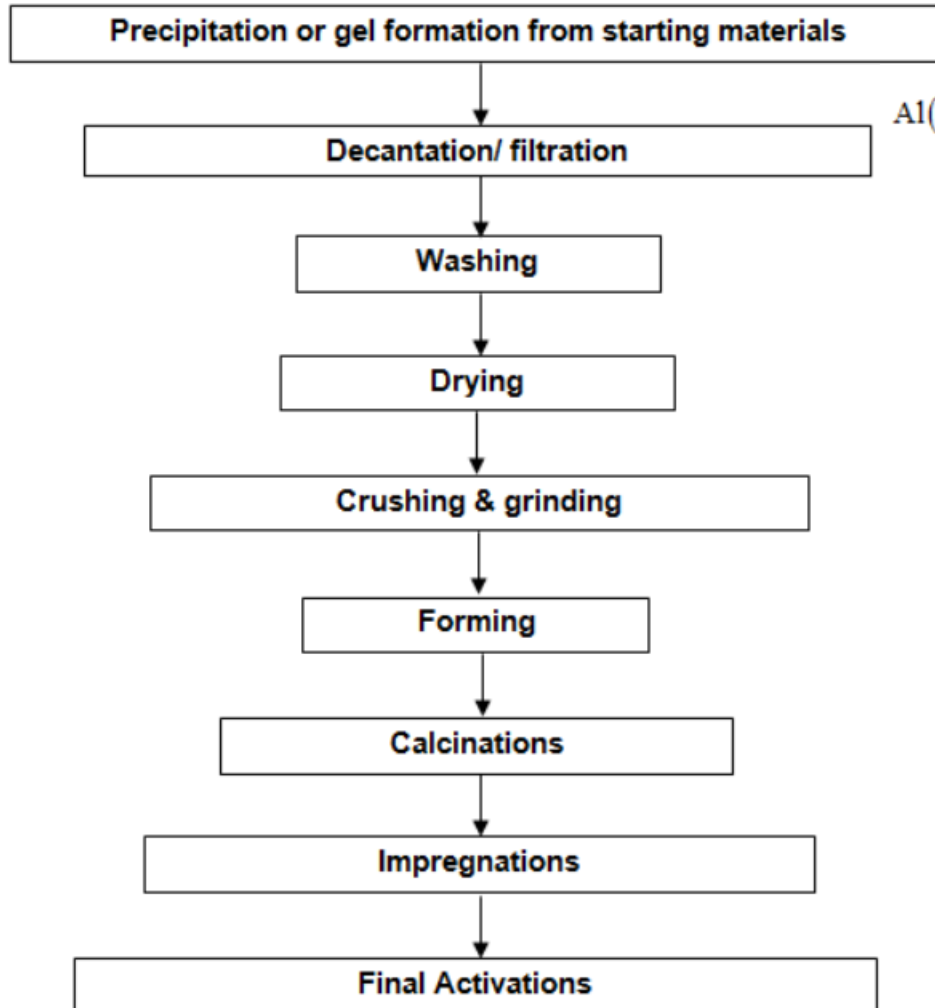
Quadlobe



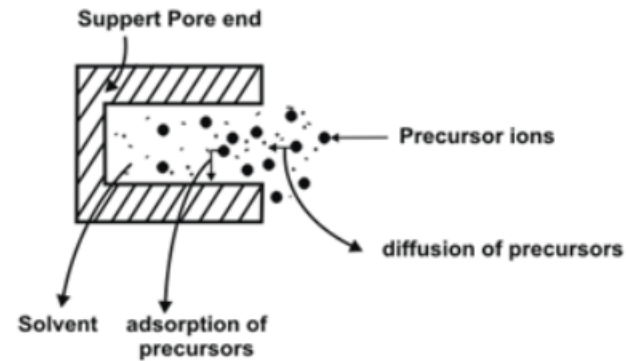
Effects of Catalyst Shapes



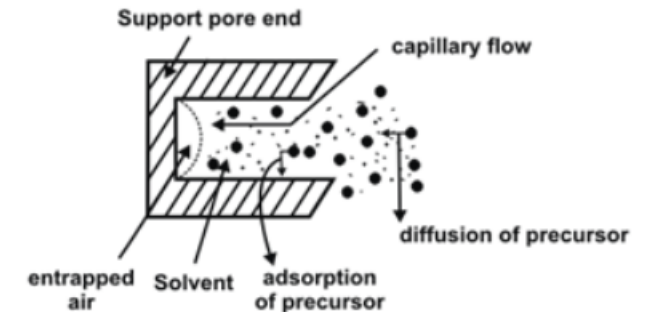
Catalyst Preparation



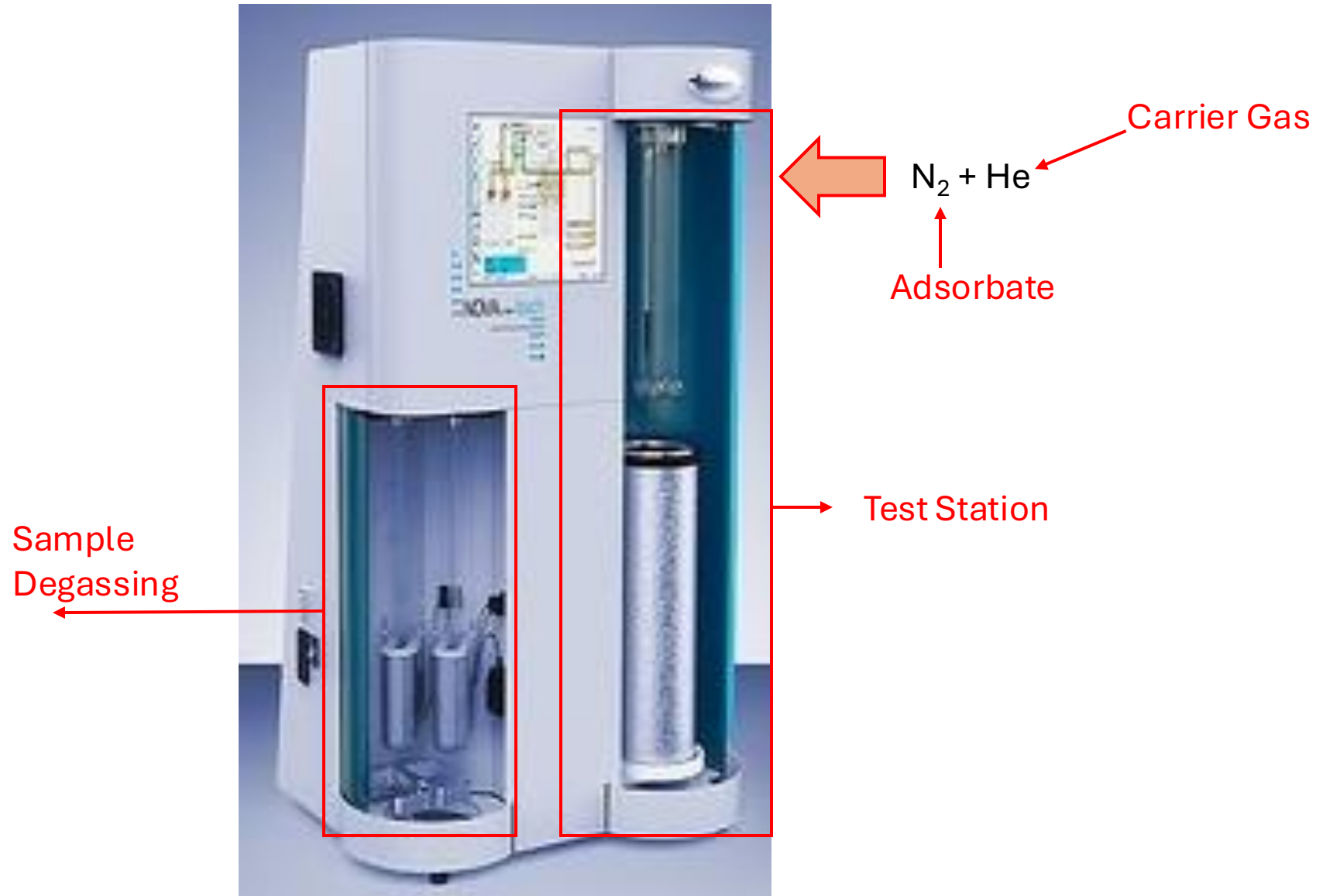
(a) Wet impregnation



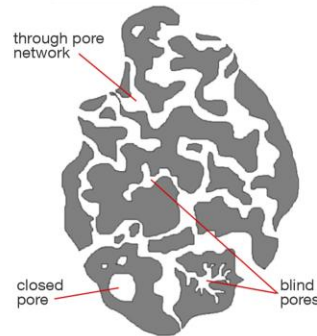
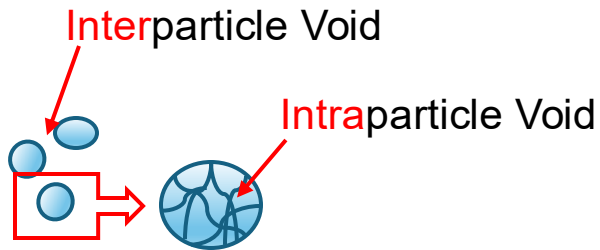
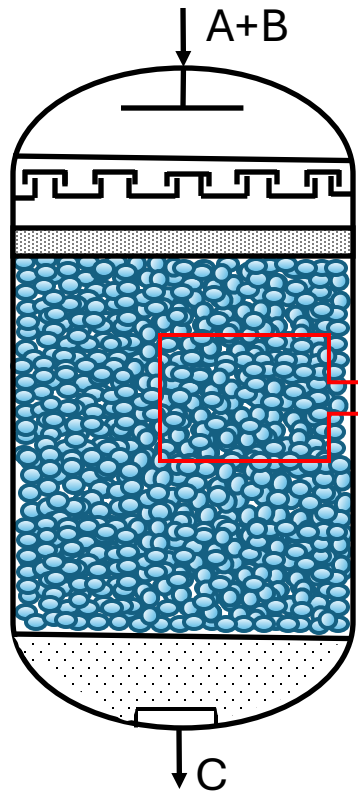
(b) Dry impregnation



BET Measurement



Density



$$\text{True Density}(\rho_s) = \frac{\text{Mass of solids}}{\text{Actual volume excluding voids}}$$

$$\text{Skeletal Density} = \frac{\text{Mass of solids}}{\text{Volume excluding interparticle voids, open pores, but including closed pores}}$$

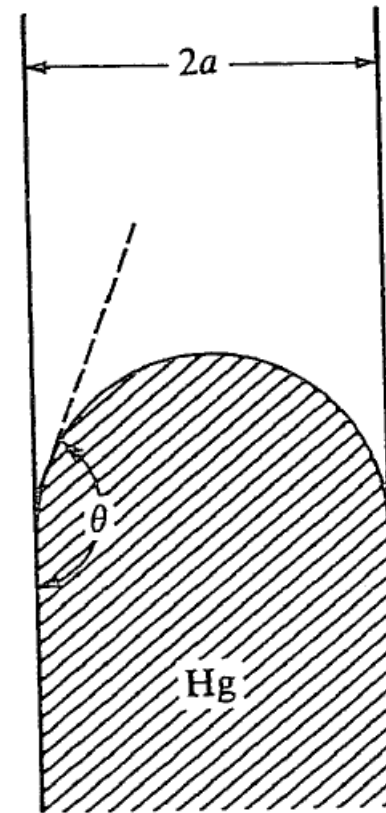
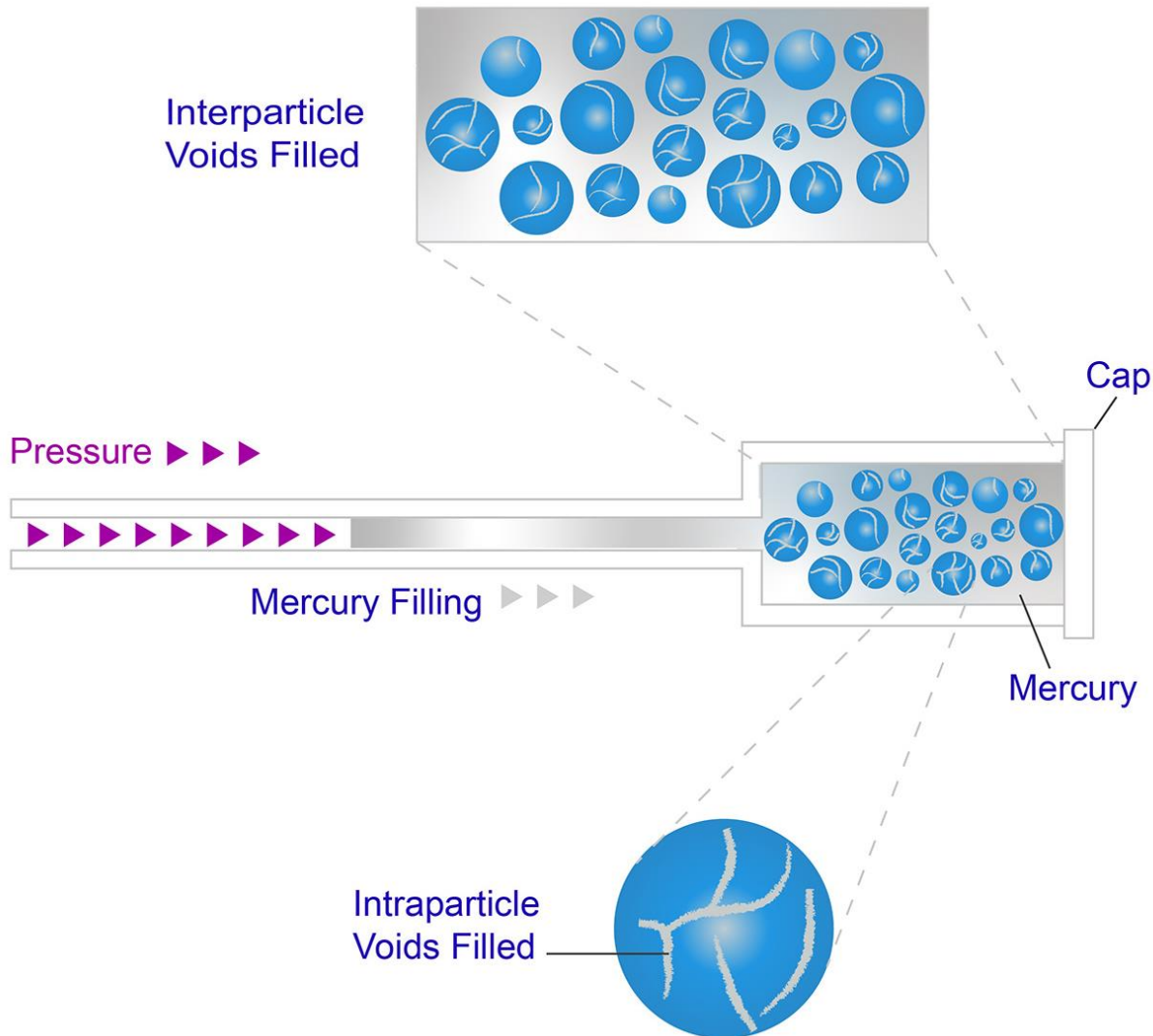
$$\text{Bulk Density} = \frac{\text{Mass of solids}}{\text{Loosely packed volume}}$$

$$\text{Transverse Axial Pressure (TAP) Density} = \frac{\text{Mass of solids}}{\text{Densely packed volume}}$$

$$\text{Pellet Density}(\rho_p) = \frac{\text{Mass of pellet}}{\text{Volume of pellet}}$$

$$\text{Pellet Porosity}(\epsilon_p) = \frac{\text{Void volume of pellet}}{\text{Total volume of pellet}}$$

Pore Volume – Mercury Intrusion Method



$$a = \frac{-2\sigma \cos \theta}{p}$$

Ritter and Drake